The main data quality issues that were encountered were the following. I also documented the resolution that I applied for each issue and the script that was applied to the python notebook Data Cleaning LIHEAP.ipynb

Data Quality Issues	Resolution
Weird column names	Use .rename to apply correct column names
Values in the Zip Code Column that are null	Use .replace to replace values titled “00nan” to pd.NA
Drop Redundant Columns	Drop the columns "Source File", "Data connected to Power BI: Click refresh to update.", "Pledge Details", "Created On", "Pledge Expiration Date"
Pledge Type has usernames that are not LIHEAP	Create a list with the known username and filted the dataframe to remove the usernames in the pledge type columns that are not liheap
Apply Correct Data types for Pledge Number, Pledge Amount and Agency Business Partner Member	Set the Pledge Number and Agency Business Partner Number columns to int64. Set pledge amount columns to float 64
Perform all date type conversions	Create new columns Date Created On and Month Year to datetime data types.
Zip Codes Standardization	Ensure that the zip codes are zero padded and have five digits, I wrote a function for this
Filter for all of the zip codes in the data that are in the san diego county	Create a list of the san diego county zips, fill in the sd zip codes with zero padding and have a zip code of five and return the records that are relevant to the analysis.

The following assumptions were made during the data cleaning process

- All records need to have a pledge type of LIHEAP
- All records need to have a zip code that is within San Diego county and set to five digits.

After standardizing the data, I checked the output for missingness as well as outliers. I wanted to assess where there were zip codes that did not generate a row within the date interval of January 2023-July 2023. What I learned was that there were 63 out of 121 zip codes, which contained at least one missing month. The zip codes of 91908, 91906, 91916, 91951 and 92014 did not capture data in six out the seven months within the data slice.

I also checked to find the rows that had outlier enrollment counts by defining the lower bound of

-11.5 and the upper bound of 24.5. I learned that there were 35 records with 16 unique zip codes that returned enrollment counts that were outliers. All of these outliers were greater than the upper bound of 24.5.

The final decision was made to retain the records with outliers as I checked for duplication and learned that there were not any duplicates, the outliers did not occur because of a data error.

I did not exclude the data where months are missing as it would created bias towards zip codes with high volume and reporting.